

FieldNotes

Physical memory for physical agency

Timestamped observations that AI agents can revisit

HACKATHON PROTOTYPE

Drone → timestamped evidence → MCP → agent. **Human controls flight.**



CONCEPT ART • NOT RECORDED EVIDENCE



observations across time (motif)

A physical decision depends on what is still true

Problem + Inspiration: 200 words maximum across slides 02-03

WHO • FOR PEOPLE OVERSEEING PHYSICAL WORK

“Is the workstation ready for the next job?”

WHAT	Earlier images may be stale.
WHY IT MATTERS NOW	Missing objects may simply be out of view.
CURRENT OPTION	Reviewing footage still takes human attention.

A capable agent needs recent evidence, not only a plausible answer.

Research: SayCan • DynaMem • Gemini Robotics (see notes)



ILLUSTRATIVE WORK SCENARIO • CONCEPT ART

Part of the surface is hidden: an unseen object is unknown, not absent.

Physical agency needs shared context

User goal → reasoning → permitted robot capability → observation → memory update → reassessment

WHAT WE OBSERVED

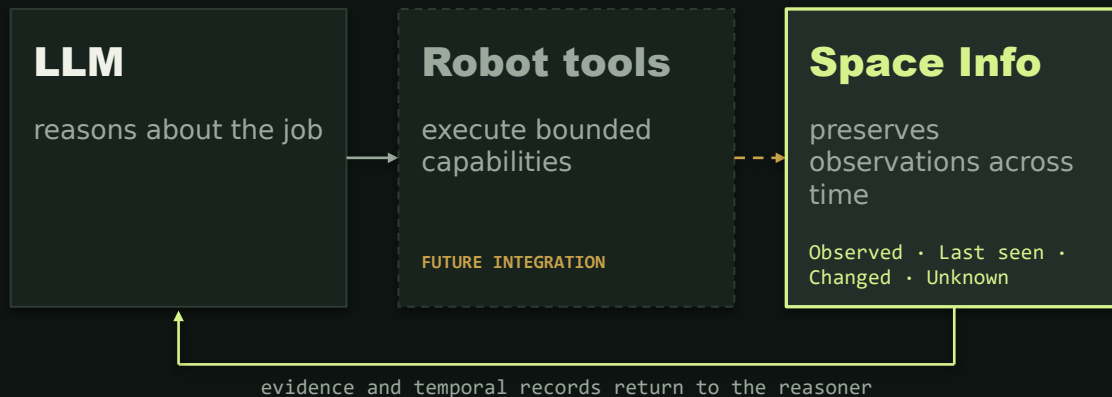
Reasoning alone answers without evidence.

WHAT IT REVEALED

The missing part is memory of what was seen, and when.

WHY IT UNLOCKS A SOLUTION

MCP lets an agent request evidence the way it requests any tool.



FieldNotes builds the temporal evidence layer

The agent can return to the evidence

Evidence-based example from one recorded session · not a verbatim transcript

WHO IT IS FOR

AI agents and the people overseeing physical work

“What changed around the workstation?”

PLACEHOLDER — INSERT ORIGINAL

Original frame 300.jpg

data/evidence/
80777bc1e2ac4fa8b2578015d93a332d/
300.jpg

Not in this checkout. Copy from the recording Mac; do not substitute generated imagery.

300.jpg 02:50:45.991 PDT · 27 Sep 2026

Person seated beside an open laptop

Camera reframed between frames: not every image difference is object movement.

PLACEHOLDER — INSERT ORIGINAL

Original frame 370.jpg

data/evidence/
80777bc1e2ac4fa8b2578015d93a332d/
370.jpg

Not in this checkout. Copy from the recording Mac; do not substitute generated imagery.

370.jpg 02:50:59.918 PDT · 27 Sep 2026

Person standing

HOW IT WORKS · THREE MCP CALLS

- 1 Search earlier observations
`search_visual_history`
- 2 Inspect original frames
`get_evidence`
- 3 Explain the supported change
`get_memory_state` / `record_memory`

Seated earlier. Standing later.

Task completion was not observed.

WHAT CHANGES: the agent cites originals instead of guessing.

148 retained evidence images

One recorded prototype session · numbers classified below

148

VERIFIED · one recorded prototype session

A record the agent can inspect after the moment has passed.

Productivity improvement has not yet been measured.

712

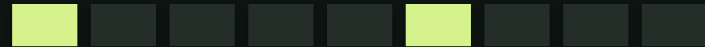
received frames

VERIFIED

140.7 s

of observed footage

CALCULATED



sampled frames retained as evidence (motif, not to scale)

HOW THE NUMBERS WERE CALCULATED

session 80777bc1... · 27 Sep 2026 · memory.sqlite3 (read-only)

Counts: rows of received frames and retained evidence.

Duration: last receipt – first receipt = 140.707 s.

The agent retrieves context when it needs it

Maximum 100 words · workflows compared, not vendors

HOW IT IS DIFFERENT

Manual review a person reconstructs the sequence

Summary alone an interpretation without the full visual context

FieldNotes MCP access to earlier originals, timestamps, and temporal records

Built today an integrated evidence workflow

Still to prove reliability across varied physical jobs

WHY IT IS POSSIBLE – AND HARD TO COPY

Originals, receipt timestamps, and temporal records are stored together and served through one MCP interface, so the agent can cite evidence rather than a summary.

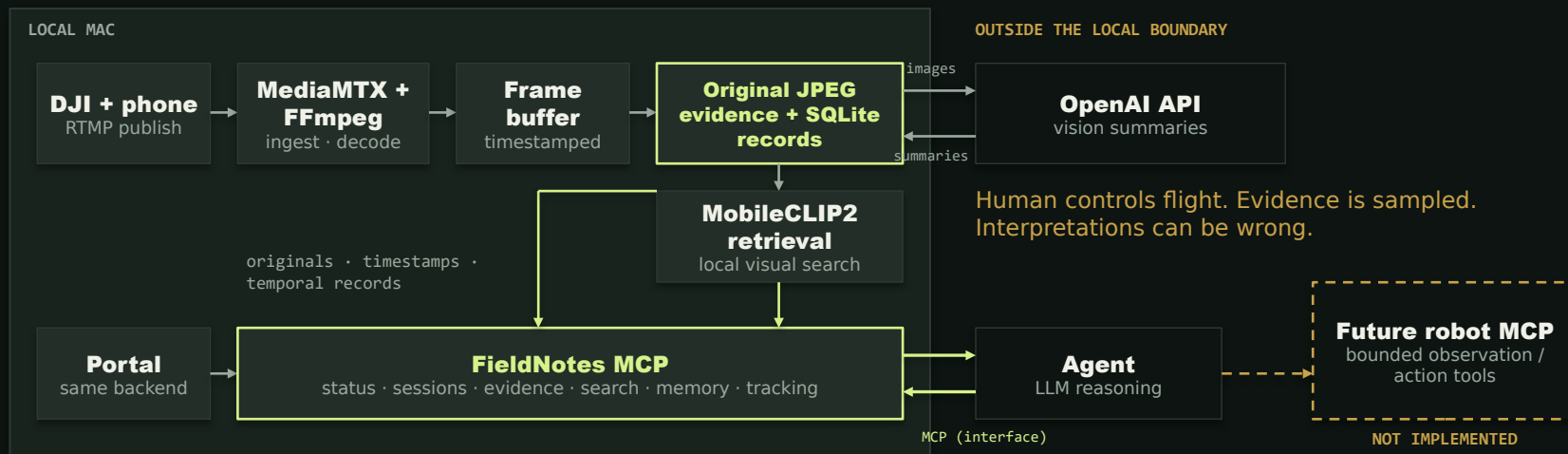
Defensibility is not a moat yet; it is accumulated workflow evaluation.

A decision can require another observation.

Requesting it through a robot tool is future work.

Observation becomes agent-accessible memory

Maximum 150 words · solid = implemented · dotted = future boundary



STACK Python 3.12 · FastMCP (Streamable HTTP) · FFmpeg · MediaMTX · SQLite · OpenAI Responses API · MobileCLIP2-S0 · EdgeTAM (optional)

Toward evidence-grounded physical agency

Planned milestones · maximum 75 words

PLANNED MILESTONES · NOT ACHIEVED RESULTS

01 NEAR TERM

**Session review and
cited process
timelines**



02 MEDIUM TERM

**Evaluate change
interpretation across
20 annotated sessions**

20 sessions = proposed target



03 LONG TERM

**Connect one bounded
robot observation tool
and verify its result**

observation first, not manipulation

The agent should know when it needs another look.